

Task 3 – Command-Line Administration Guide

Tech Career Accelerator Week 2
Deliverable

1. Purpose

This guide provides system administrators with a concise reference for the Linux command-line instructions most frequently required during day-to-day server and workstation administration. Each command is accompanied by a description, common syntax, and a practical example relevant to the Tech Career Accelerator environment.

2. Command Reference

pwd – Print Working Directory

Displays the absolute path of the current working directory. Useful for confirming location before executing file-system operations.

Syntax:

```
pwd
```

Example:

```
$ pwd /home/administrator
```

ls – List Directory Contents

Lists files and directories within a specified path. The `-la` flag displays hidden files and detailed metadata including permissions, ownership, and modification dates.

Syntax:

```
ls [options] [path]
```

Example:

```
$ ls -la /Company/Finance total 48 drwxr-x--- 3 root finance_grp 4096 May 10 09:00 .  
drwxr-xr-x 10 root root 4096 May 10 08:55 ..
```

cd – Change Directory

Navigates to the specified directory. Use `cd ..` to move up one level, or `cd ~` to return to the current user's home directory.

Syntax:

```
cd [path]
```

Example:

```
$ cd /Company/HR $ pwd /Company/HR
```

mkdir – Make Directory

Creates one or more new directories. The -p flag creates all intermediate parent directories as required without returning an error if the path already exists.

Syntax:

```
mkdir [options] directory_name
```

Example:

```
$ mkdir -p /Company/Sales/Archive $ ls /Company/Sales Archive Reports Templates Working
```

touch – Create or Update File Timestamp

Creates an empty file at the specified path or, if the file already exists, updates its access and modification timestamps.

Syntax:

```
touch filename
```

Example:

```
$ touch /Company/Public/welcome.txt $ ls -l /Company/Public/ -rw-r--r-- 1 admin all_staff  
0 May 10 10:15 welcome.txt
```

chmod – Change File Mode (Permissions)

Modifies read, write, and execute permissions for the owner, group, and others. Permissions may be expressed in octal notation or symbolic form.

Syntax:

```
chmod [options] mode file
```

Example:

```
$ chmod 750 /Company/Finance # Owner: full (7), Group: read+execute (5), Others: none (0)
```

chown – Change File Owner and Group

Transfers ownership of a file or directory to a specified user and/or group. The -R flag applies the change recursively to all contents.

Syntax:

```
chown [options] user[:group] file
```

Example:

```
$ chown -R root:finance_grp /Company/Finance $ ls -ld /Company/Finance drwxr-x--- 4 root  
finance_grp 4096 May 10 09:00 /Company/Finance
```

ps – Process Status

Reports a snapshot of currently running processes. The aux flags display all processes for all users in a user-readable format, including those without a controlling terminal.

Syntax:

```
ps [options]
```

Example:

```
$ ps aux | grep apache2 root 1234 0.0 0.1 apache2 -k start
```

top – Real-Time Process Monitor

Provides a dynamic, continuously updated view of system resource utilisation, including CPU usage, memory consumption, and the processes consuming the most resources. Press q to exit.

Syntax:

```
top
```

Example:

```
$ top top - 10:22:01 up 3 days, 2:14 Tasks: 142 total, 1 running, 141 sleeping %Cpu(s): 3.1 us, 0.8 sy, 0.0 ni, 95.6 id
```

apt install – Install Software Packages

Installs one or more packages from the configured APT repositories on Debian/Ubuntu-based systems. Always run apt update beforehand to refresh the package index.

Syntax:

```
sudo apt update && sudo apt install package_name
```

Example:

```
$ sudo apt update $ sudo apt install openssh-server Reading package lists... Done The following NEW packages will be installed: openssh-server
```

3. Best Practices

- Always verify the current working directory with pwd before executing destructive or recursive operations.
- Use the -i (interactive) flag with rm and mv to prompt for confirmation before overwriting or deleting files.
- Restrict use of sudo to required tasks only; do not operate as root for routine administration.
- Pipe command output through less for long listings (e.g., ls -la | less) to enable scrollable review.
- Document all non-standard permission changes in the IT change log immediately after execution.